ONLINE ASSIGNMENT TRACKER

Project DocumentationPROJECT DOCUMENTATION

Prepared By

Project Information	Details
Project Name	Online Assignment Tracker
Application Type	Web Application
Backend	Python with Django
Database	SQLite
Frontend	HTML5, CSS3, Bootstrap 5, JavaScript, Django Templates
Version Control	Git and GitHub
Assignment Allocation	Course and Batch Wise
User Roles	Student and Teacher

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A web-based assignment management system for students and teachers

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1. Introduction

Online Assignment Tracker is a web-based application developed to simplify the process of creating, assigning, submitting, reviewing and grading academic assignments. It provides separate workflows for teachers and students.

The application is developed using Python and Django for the backend, SQLite for data storage, and HTML, CSS, Bootstrap and JavaScript for the user interface. Django Templates connect frontend pages with backend data and application logic.

2. Problem Statement

Managing assignments manually through paper, messages or separate files can make it difficult to track deadlines, student submissions, marks and feedback. Teachers may also need to assign the same assignment to many students in a particular course and batch.

The proposed system provides a centralized platform where assignment activities can be managed in a structured and trackable manner.

3. Objectives

- Provide separate registration and login workflows for students and teachers.
- Allow teachers to create assignments according to course and batch.
- Automatically create pending submission records for students in the selected course and batch.
- Allow students to view assignments relevant to their course and batch.
- Allow students to upload assignment files online.
- Allow teachers to view submissions and provide marks and feedback.
- Maintain assignment and submission information in a structured database.
- Provide a responsive interface using Bootstrap.

4. Scope of the Project

- User registration and authentication.
- Teacher subject selection.
- Student course and batch selection.
- Course and batch wise assignment creation.
- Assignment viewing and details.
- Student file submission.
- Teacher submission review.
- Marks and feedback management.
- Submission status tracking.

5. Technologies Used

Technology	Purpose
Python	Main programming language.
Django	Web framework for backend and server-side application.

SQLite	Database for application data.
HTML5	Web page structure.
CSS3	Styling.
Bootstrap 5	Responsive UI components.
JavaScript	Client-side interactions.
Django Templates	Display backend data in web pages.
Django ORM	Database interaction through Python models.
Git	Version control.
GitHub	Remote source-code repository.

6. System Requirements

Software Requirements

- Windows, Linux or macOS.
- Python 3.x.
- Visual Studio Code or another code editor.
- Chrome, Edge or Firefox.
- Git for version control.

Hardware Requirements

- Basic computer/laptop.
- Minimum 4 GB RAM recommended.
- Sufficient storage for project and uploaded files.

7. System Architecture

The browser sends requests to Django views. Views use Django models and the ORM to read or update SQLite data, and Django Templates generate the HTML response.

Layer	**Main Components**
Frontend	HTML, CSS, Bootstrap, JavaScript, Django Templates
Backend	Django Views, Forms, URL routing, validation and business logic
Data Layer	Django Models, Django ORM and SQLite
File Storage	Media directory for uploaded assignment files

8. Modules / Components

Module	**Entity / Purpose**
User Management	User / Profile
Student Management	Profile
Teacher Management	Profile
Assignment Management	Assignment
Submission Management	Submission
Grading & Feedback	Submission
Authentication	User / Profile

9. User Roles and Responsibilities

Student

- Register with course and batch.
- Log in.
- View assignments matching course and batch.
- Open assignment details.
- Upload assignment files.
- View marks and feedback.

Teacher

- Register with subject.
- Log in.
- Create assignments.
- Select target course and batch.
- View assigned students and submissions.
- Enter marks and feedback.

10. Course and Batch Wise Assignment Flow

When a teacher creates an assignment, a target course and batch are selected. The system finds students with the same course and batch and creates a pending submission record for each matching student.

1. Teacher logs in.
2. Teacher opens Create Assignment.
3. Teacher enters title, description and due date.
4. Teacher selects target course and batch.
5. Teacher's subject is associated with the assignment.
6. System identifies matching students.
7. Pending submission records are created.
8. Matching students see the assignment.

11. Database Design / Entities

The project uses Django models with SQLite. The main entities are Profile, Assignment and Submission. Django's built-in User model is used for authentication.

Entity	Important Information
User	Authentication account used for login.
Profile	Role, subject, course and batch.
Assignment	Title, description, subject, target course, target batch, assigned date, due date and teacher.
Submission	Student, assignment, file, submission time, status, marks and feedback.

Course and batch are represented through predefined choice fields rather than separate database tables.

12. Frontend Design

The frontend uses Django Templates, HTML5, CSS3, Bootstrap 5 and JavaScript. Main pages include:

- Home page
- Registration page
- Login page
- Student dashboard
- Assignment detail page
- Teacher dashboard
- Create assignment page
- Teacher assignment detail page
- Grade submission page

Bootstrap is used for responsive forms, cards, tables, buttons and navigation elements.

13. Backend Design

Django manages routing, authentication, forms, models, validation and business logic.

Component	Responsibility
models.py	Defines Profile, Assignment and Submission.
forms.py	Registration, assignment, submission and grading forms.
views.py	Requests, authorization, allocation, submissions and grading.
urls.py	Maps URLs to views.
admin.py	Registers models for Django administration.
migrations/	Database schema migrations.
settings.py	Project configuration, templates, static/media and database settings.

14. File Upload and Submission Flow

9. Student opens an assignment.
10. Student selects a file.
11. Backend validates the submission and deadline.
12. File is stored through media/file handling.
13. Submission record is updated.
14. Teacher can view the submitted assignment.

15. Grading and Feedback Flow

15. Teacher opens an assignment and views assigned students.

16. Teacher reviews a submitted file.
17. Teacher enters marks, feedback and status.
18. Evaluation is saved.
19. Student can view marks and feedback.

16. Authentication and Validation

- Django authentication is used for login and logout.
- Registration collects role-specific information.
- Teacher registration includes subject selection.
- Student registration includes course and batch selection.
- Forms validate input.
- Students are restricted to matching course and batch assignments.
- Teacher grading is available when a submission file exists.
- Submission deadline is validated.

17. Project Structure

```
online-assignment-tracker/
│
├── Backend/
│   ├── assignment_tracker/
│   ├── tracker/
│   ├── manage.py
│   └── requirements.txt
│
├── Frontend/
│   ├── templates/
│   └── static/
│
├── README.md
└── .gitignore
```

18. Installation and Setup

Open the project in Visual Studio Code and open a terminal in the project root.

```
cd "C:\Users\<username>\Downloads\online_assignment_tracker_course_batch"

python -m venv venv

venv\Scripts\activate

pip install -r Backend\requirements.txt

python Backend\manage.py migrate
```

If the virtual environment already exists, activate it directly.

19. How to Run the Project

Start the server: `python Backend\manage.py runserver`

Open: `http://127.0.0.1:8000/`

Stop the server with Ctrl + C.

20. Testing

Test Area	Expected Result
Student registration	Account is created with course and batch.
Teacher registration	Account is created with subject.
Login	Correct user can access the application.
Assignment creation	Teacher creates assignment for selected course and batch.
Assignment allocation	Matching students receive pending submission records.
Student dashboard	Only matching assignments are shown.
File submission	Student can upload an assignment file.
Deadline validation	Late submissions are restricted according to due date.
Grading	Teacher can enter marks and feedback.
GitHub repository	Frontend and Backend are organized; ignored files are not tracked.

21. Advantages

- Centralized assignment management.
- Reduces manual assignment distribution.
- Course and batch wise allocation.
- Easy online file submission.
- Submission status tracking.
- Marks and feedback stored with submissions.
- Responsive Bootstrap interface.
- Simple SQLite database for local projects.

22. Limitations

- Primarily designed for local or small-scale use.
- SQLite may not be ideal for large multi-user deployment.
- The current version does not implement a separate REST API layer.
- Uploaded files require suitable storage configuration for production.
- Course and batch values are maintained as predefined choices.

23. Future Enhancements

- Add REST API using Django REST Framework.

- Add email notifications and deadline reminders.
- Add search, filtering and sorting.
- Add downloadable marks/submission reports.
- Use PostgreSQL for production.
- Add cloud file storage.
- Add administrator dashboard.
- Add assignment and performance analytics.

24. Conclusion

Online Assignment Tracker provides a structured solution for managing academic assignments between teachers and students. The system combines Django backend functionality, a responsive Bootstrap-based frontend and SQLite database storage. Course and batch wise allocation makes assignment distribution easier, while students get a dedicated workflow for viewing, submitting and receiving feedback.

25. GitHub Repository

https://github.com/Sakshi-shahu/online-assignment-tracker

— End of Project Documentation —